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CHEMISTRY NOW: Blowing Up Balloons With Yeast (Grades 5-8)



Objective:

Students will make observations of solids, liquids and gases. Students will conduct an experiment to visually demonstrate the formation of a gas by yeast during a chemical change. Students will investigate the effects of changing variables on this chemical change. Students will be able to describe physical properties of a solid, liquid and gas. Students will be able to describe the formation of a gas during a chemical change.

Introduction Notes:

CHEMISTRY NOW: Blowing Up Balloons With Yeast

Subject Area: Chemistry

Grade Level: Middle School Chemistry

Lesson Title: Blowing Up Balloons With Yeast

National Science Standards:

Science Teaching Standards - A, B, & E

Assessment Standard - C

Physical Science Standards:

5-8 Properties and Changes of Properties

Life Science Standards:

5-8 Structure and Function in Living Systems

Suggested Prior Knowledge: concepts of solid, liquid, gas, chemical changes; skills of observation

Purpose: To allow students to observe a chemical change and investigate what variables affect this chemical change.

Key Vocabulary:

yeast – any of various unicellular fungi of the genus *Saccharomyces*, especially *S.cerevisiae*, reproducing by budding and from ascospores and capable of fermenting carbohydrates

solid – matter with definite shape and volume, consisting of particles fixed in location

liquid – matter with definite volume that can take the shape of its container, consisting of particles close together that can flow past each other

gas – matter that takes the shape and volume of its container, consisting of particles that are very far apart and moving freely and rapidly

substance – an element or compound, matter with uniform chemical composition

chemical change – change in which a substance or substances are changed into a new substance or substances

physical property – property of matter that can be observed without changing the identity of substance(s), such as color, texture, state of matter

Objectives:

1. Students will make observations of solids, liquids and gases
2. Students will conduct an experiment to visually demonstrate the formation of a gas by yeast during a chemical change
3. Students will investigate the effects of changing variables on this chemical change
4. Students will be able to describe physical properties of a solid, liquid and gas
5. Students will be able to describe the formation of a gas during a chemical change

Materials: (per each group of 2-3 students)

- small, clear plastic soda bottle
- packet of yeast
- tsp. of sugar
- some warm water
- small balloon

Procedure:

A. Background information: Yeasts are unicellular organisms that are best known as the micro-organisms that make bread rise. In the absence of oxygen gas, Yeasts carry out fermentation and convert sugar (glucose) into another organic compound (lactic acid or ethyl alcohol) and carbon dioxide gas which makes the bubbles in bread and causes the bread to rise. Fermentation is an example of a chemical change that forms a gas. In the process of fermentation the sugar molecules are broken down to release energy for the yeasts to use to live and carbon dioxide is released as a waste product. Yeast cells provide enzymes (biological catalysts) that allow fermentation to take place. Yeasts are also used to make beer and wine since alcohol can also be a product of fermentation.

B. Introduce the basic experimental procedure to students. Discuss with students how yeast is used to make carbon dioxide gas in such foods as bread. Original demonstration of the chemical change can be carried out by the teacher or by small groups of students.

Begin with a lead-in question and follow up:

a. What makes bread rise?

- Carbon dioxide gas bubbles in the dough cause the bread to rise. This gas is produced by yeast (a leavening agent) during the rising and baking process. Yeasts carry out fermentation using sugars to make energy and carbon dioxide gas.

b. What is a leavening agent?

- Yeast or other substance used to make gas during baking

c. What state of matter is this substance/mixture?

- Yeast is solid, water is liquid, carbon dioxide gas is formed

d. What are some physical properties of this substance/mixture?

- Physical properties include color, texture, state of matter, quantity/amount

e. How does this substance/mixture compare to others in this activity?

f. What do we know about chemical changes?

- Chemical changes occur when one or more substances change into a new substance or new substance

g. How do you know a gas is being formed?

- Bubbling occurs

C. Be certain that lab safety equipment such as goggles are used, and safety protocols are followed.

D. The basic experiment is outlined below:

1. Obtain a small, clear plastic soda bottle and a packet of yeast

2. Empty the packet of yeast into the soda bottle.

3. Observe the yeast and record your observations in Data Table 1

4. Add some warm water to the soda bottle and swirl to mix the two substances.

5. Observe the mixture in the bottle and record your observations in Data Table 1

6. Add a tsp. of sugar to the mixture in the soda bottle; stretch out the balloon by blowing it up a few times, then place it over the top of the soda bottle.

7. Observe both the mixture in the soda bottle and the balloon for about 20 minutes and record your observations in Data Table 1.

E. Work with students after this initial investigation to develop experiments to examine ways to make more gas with this basic chemical change. Lead the students toward a technically correct experiment that will allow them to test the effects of some variables such as: changes in the water temperature, changes in the room temperature, changes in the size of the soda bottle, and changes in the "food" for the yeast (honey, sugar substitute, syrup) on the amount of gas formed. Students may even try other kitchen chemicals to try to produce more gas and blow up the balloon the most.

F. The teacher may want to make this a multi-day lesson, with the basic lab procedure completed the first day, and follow-up experiments on a second day. This allows time to gather any special equipment (such as a source of hot and cold water) or materials (other "foods").

G. Students may want to use string to measure the circumference of the inflated balloons to compare size more accurately.

Additional Resources:

<http://www.newton.dep.anl.gov/askasci/mole00/mole00195.htm>

<http://www.answers.com/topic/yeast>

http://wiki.answers.com/Q/How_does_yeast_give_off_gas

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Student Worksheet for: Blowing Up Balloons With Yeast

Name: _____ Date: _____

Materials:

- small, clear plastic soda bottle
- packet of yeast
- tsp. of sugar
- some warm water
- small balloon

Safety: Use care with all equipment and substances; wear goggles

Basic Procedure:

1. Obtain a small, clear plastic soda bottle, some warm water, some sugar and a packet of yeast
2. Design an experiment to investigate ways to make more gas using the basic procedure demonstrated. You might want to think about the following questions:
3. Does the water need to be warm? Does warmer or cooler water make more gas?
4. Do other “foods” work for the yeast?
5. Does a bigger bottle help?
6. Can other kitchen substances do this too?
7. Be certain to carefully record all steps of your procedure and all observations.

Student Hypothesis: _____

Student Procedure: _____

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

Substance(s)	Observations
Dry yeast	
Yeast and water mixture	
Yeast, water and sugar mixture	
Balloon	

Other Observations:

Conclusion:

What factors affect the process of fermentation carried out by yeast?

How do these affect fermentation?

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30 Rockefeller Plaza New York, NY 11102 (877) NBC - 7502



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